Deep Learning & Neural Network Architectures

A Comprehensive Engineering Handbook

By Dr. K. Srinivas & Prof. R. V. Rao

Department of Computer Science & Engineering

Published by NRI Institute of Technology Academic Press

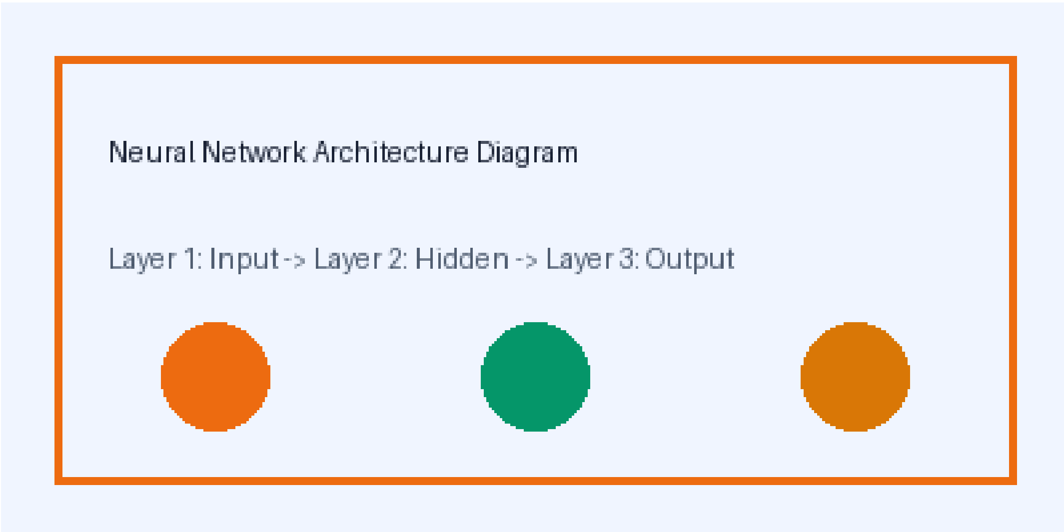
ISBN: 978-0-13-468599-1

Chapter 1: Foundations of Artificial Intelligence

1.1 Historical Evolution of Neural Models

Artificial intelligence and computational neural networks have evolved from basic perceptrons into multifacet

Key Takeaway: Gradient descent optimization is fundamental to modern deep learning convergence.



· Biological vs Artificial Neuron Architectures
· Mathematical Formulations of Activation Functions
· Backpropagation and Loss Surface Exploration

Chapter 2: Convolutional and Recurrent Architectures

2.1 Spatial Feature Hierarchies

Convolutional Neural Networks utilize local receptive fields, shared weights, and spatial pooling to extract in

2.2 Sequence Modeling with Transformers

Self-attention mechanisms allow sequence modeling without recurrence constraints, facilitating massive par